

GRAFENWOHR, Germany

WMO: 106870

Lat: 49.699N

Lon: 11.940E

Elev: 1363

StdP: 13.99

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.3	14.4	3.3	6.8	9.9	9.4	9.3	17.8	18.4	38.8	16.1	37.2	1.5	320	0.524

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.1	87.9	66.8	83.9	65.6	80.3	64.9	69.4	82.4	67.5	79.9	66.0	77.2	5.3	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.7	99.7	72.5	63.9	93.5	70.7	62.3	88.3	69.6	34.2	82.5	32.7	80.2	31.5	77.0	79.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.6	12.7	11.3	DB	1.9	93.8	7.1	3.2	-3.2	96.1	-7.4	98.0	-11.3	99.7	-16.5	102.0	(4)
(5)			WB	0.6	72.8	6.7	2.4	-4.2	74.6	-8.1	75.9	-11.8	77.3	-16.7	79.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.3	30.2	32.8	38.8	46.8	54.7	60.9	64.9	63.4	55.2	46.9	38.6	33.5
	DBStd	13.76	8.28	8.16	6.60	7.30	6.82	6.61	5.96	5.86	6.00	6.71	6.29	6.89
	HDD50	2634	614	482	350	145	30	1	0	0	17	140	343	512
	HDD65	6632	1079	902	813	547	327	158	78	100	299	561	791	977
	CDD50	1651	0	0	2	48	175	329	462	416	172	45	2	0
	CDD65	173	0	0	0	0	7	36	75	51	4	0	0	0
	CDH74	2631	0	0	0	26	145	565	1022	766	106	1	0	0
	CDH80	832	0	0	0	2	22	178	370	243	17	0	0	0
Wind	WSAvg	3.9	4.8	4.2	5.1	4.3	3.9	3.8	3.7	3.1	3.0	3.1	4.0	4.2
Precipitation	PrecAvg	30.3	2.6	2.1	2.1	1.7	2.7	3.1	3.2	2.7	2.5	2.3	2.6	2.8
	PrecMax	44.3	6.0	5.5	5.6	3.8	6.1	5.4	5.1	6.5	6.2	7.9	5.9	7.3
	PrecMin	20.7	0.3	0.4	0.4	0.3	0.7	0.7	0.7	0.8	0.7	0.6	0.0	0.5
	PrecStd	4.8	1.3	1.3	1.3	0.8	1.4	1.2	1.2	1.3	1.1	1.4	1.3	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.6	55.7	65.0	77.2	82.7	91.0	93.3	92.0	82.4	71.6	58.3	51.5
		MCWB	47.8	46.9	52.0	57.5	65.0	68.4	67.7	67.4	63.9	58.8	53.0	48.0
	2%	DB	47.2	51.5	60.4	71.9	78.5	85.7	88.1	86.2	77.0	65.8	53.9	48.0
		MCWB	44.6	46.2	48.1	55.5	63.3	66.4	66.6	66.4	62.5	57.3	50.6	46.0
	5%	DB	44.4	48.0	55.1	67.9	73.7	80.6	84.1	82.1	72.0	61.1	51.5	44.9
		MCWB	42.2	43.6	46.3	54.2	59.6	65.0	65.5	66.0	60.5	55.5	48.8	43.3
	10%	DB	41.1	44.6	50.1	62.8	69.7	76.2	79.7	78.3	67.9	57.6	48.2	42.8
		MCWB	39.5	41.8	44.4	51.3	57.7	62.7	64.5	64.1	58.6	53.5	46.6	41.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.5	49.7	53.5	59.8	67.8	71.4	70.6	72.0	66.5	61.0	54.1	48.6
		MCDB	50.2	53.8	60.9	72.5	78.6	84.4	83.1	81.3	75.9	67.0	56.1	50.0
	2%	WB	45.0	46.6	49.8	57.3	64.5	68.3	68.6	69.2	63.9	58.6	51.3	46.2
		MCDB	46.8	50.1	56.6	69.0	75.6	80.7	82.4	81.3	74.3	63.6	53.5	47.6
	5%	WB	42.2	44.0	47.5	54.9	61.5	65.8	67.2	67.3	61.9	56.5	48.9	43.3
		MCDB	43.6	46.9	53.2	65.8	71.0	77.4	79.9	79.3	70.6	60.9	50.9	44.3
	10%	WB	39.4	41.6	45.1	52.5	58.8	63.6	65.7	65.4	60.0	54.1	46.6	41.6
		MCDB	40.7	44.2	49.4	62.1	67.0	74.1	76.9	75.8	66.7	57.6	48.1	42.6
Mean Daily Temperature Range	5% DB	MDBR	10.1	14.0	17.0	22.6	21.9	22.2	23.1	23.3	20.8	17.2	10.9	8.9
		MCDBR	12.2	20.8	28.4	33.1	30.2	31.5	32.8	31.6	28.4	23.3	16.4	11.6
		MCWBR	10.3	15.0	18.0	17.9	15.6	14.6	12.8	13.9	15.5	15.8	13.2	9.7
	5% WB	MCDBR	11.2	17.7	24.1	29.9	26.7	28.2	27.0	27.3	25.6	22.0	15.1	11.1
MCWBR		10.0	13.8	16.4	17.1	15.1	14.1	12.1	14.3	14.7	15.2	12.9	9.9	
Clear-Sky Solar Irradiance	taub	0.298	0.320	0.358	0.393	0.399	0.399	0.410	0.403	0.374	0.349	0.324	0.295	
	taud	2.412	2.365	2.329	2.270	2.285	2.310	2.276	2.316	2.375	2.436	2.460	2.427	
	Ebn at Noon	239	260	268	270	273	273	267	264	260	246	227	223	
	Edh at Noon	20	27	33	39	40	39	40	37	31	25	19	17	
All-Sky Solar Radiation	RadAvg	268	533	868	1331	1570	1728	1641	1433	1033	616	296	208	
	RadStd	48	84	115	174	182	144	166	141	140	75	36	25	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (55 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.54	+0.68	N/A	N/A	N/A	+36

Nomenclature: See separate page